Extracting Accountability Graphs from Regulatory Text: A Structural Analysis of FDA Warning Letter Corpus

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Abstract Accountability in regulatory investigations is not merely a legal designation — it is a relational structure produced through documentation practices. I present a graph-based framework for extracting accountability relationships from regulatory text and representing them as directed graphs. Applied to an FDA Warning Letter corpus (N=50, 2016–2025, 21 CFR 210/211), the framework extracts 320 accountability triples across three relation types (VIOLATES, INVOLVES, ATTRIBUTED_TO), constructs case-level accountability graphs, and computes structural metrics including centrality, openness score, and cluster membership. Graph analysis reveals a star-topology accountability structure with Firm as the dominant node (in-degree centrality = 0.444). A composite accountability openness score exhibits significant negative correlation with premature convergence risk (Pearson $r = -0.431$, $p = 0.0018$; Spearman $\rho = -0.742$, $p < 0.001$), providing cross-framework validation of structural accountability concentration. Cases cluster into three structural types: Open, Standard, and Extreme convergence.

Keywords: accountability graphs, relation extraction, regulatory NLP, FDA Warning Letters, network analysis, premature convergence

Introduction

Accountability in high-stakes organizational settings is not simply assigned — it is constructed through sequential documentation practices that link actors, actions, and regulatory obligations. When these linkages are represented computationally, they reveal structural properties that cannot be observed through text analysis alone: which actors occupy central positions, how

violations cluster across regulatory domains, and whether accountability structures are open or concentrated.

Prior work on regulatory text analysis has focused primarily on classification tasks (2) or compliance detection, without examining the relational structure of accountability as a graph-theoretic problem. I address this gap by presenting a framework for extracting accountability graphs from regulatory documents and analyzing their structural properties.

This paper makes three contributions:

- I formalize accountability relation extraction as a triple extraction task over regulatory text, producing directed graphs with typed edges.
- I design a composite accountability openness score combining violation count, CFR diversity, and text diversity.
- I validate the framework against a PAC detection framework (1), demonstrating convergent validity across two independent computational methods.

Related Work

Relation Extraction

Relation extraction involves identifying typed relationships between entities in unstructured text and representing them as structured triples (4). It serves as a foundation for knowledge graph construction, question answering, and document understanding across biomedical, legal, and scientific domains. I extend this methodology to the regulatory enforcement domain, defining accountability-specific relation types grounded in the logic of FDA Warning Letter documents.

Network Analysis and Centrality

Social network analysis provides formal methods for analyzing relational structures among entities (5). Centrality measures — including in-degree, betweenness, and PageRank — quantify the structural importance of nodes within a network. I apply these methods to accountability

graphs, treating regulatory actors and CFR sections as nodes and enforcement relationships as typed directed edges.

Computational Accountability

Recent work has operationalized accountability as a computational construct (1; 2; 3), but has not examined its graph-level structural properties. I extend this program by treating accountability not as a label or score, but as a relational network embedded in document structure.

Framework

Relation Type Definition

I define three accountability relation types derived from the structure of FDA Warning Letters:

- **VIOLATES**: links a responsibility subject to a CFR regulatory section
- **INVOLVES**: links a violation instance to its textual description
- **ATTRIBUTED_TO**: links a violation instance to its assigned responsibility subject

These relations capture the three core accountability questions embedded in regulatory enforcement text: who is responsible, what regulation was violated, and what the violation involved.

Graph Construction

For each violation record, I extract a unique violation node (CaseID + ViolationID), a responsibility subject node, a CFR node, and a violation text node. Directed edges are added according to the relation types defined above, producing a multi-relational directed graph $G = (V, E)$ where each edge $e \in E$ carries a relation type label.

Structural Metrics

I compute three graph-level metrics per case. *In-degree centrality* measures how many accountability arrows point to a given node, capturing responsibility concentration. *Betweenness centrality* measures structural bridge roles — nodes that connect otherwise disconnected parts of the accountability graph (5). *Accountability openness score* is a composite metric:

$$o = w_1 \cdot \tilde{n}_v + w_2 \cdot \tilde{n}_c + w_3 \cdot \tilde{d}_t \tag{1}$$

where $\tilde{n}_v$ is normalized violation count, $\tilde{n}_c$ is normalized unique CFR count, $\tilde{d}_t$ is normalized text diversity (TF-IDF based), and $w_1 = 0.3$, $w_2 = 0.4$, $w_3 = 0.3$.

Experiments

Dataset

I apply the framework to the FDA Warning Letter corpus (N=50, 2016–2025, 21 CFR 210/211), comprising 160 violation records across 50 finished pharmaceutical manufacturing cases (2). From this corpus, I extract 320 accountability triples (160 VIOLATES + 160 INVOLVES), constructing a directed graph with 359 nodes and 363 edges.

Graph Structure Analysis

Table 1 reports global graph statistics.

Table 1: Accountability Graph Statistics

Metric	**Value**
Total nodes	359
Total edges	363
Relation types	3
Unique CFR sections	68
Dominant node in-degree centrality	0.444
Dominant node betweenness centrality	0.052

Centrality Analysis

Figure 1 shows the full accountability graph. Figures 2 and 3 report centrality distributions. Firm exhibits in-degree centrality of 0.444 — receiving 44% of all incoming accountability edges — and dominates betweenness centrality at 0.052, with all other nodes near zero. The zero clustering coefficient across all nodes confirms the star-shaped topology: responsibility flows from violations to a single subject without forming triangular accountability relationships.

Openness Score and Cross-framework Validation

Figure 4 shows openness scores across all 50 cases. Figure 5 presents the cross-framework validation scatter plot. Accountability openness score exhibits significant negative correlation with PAC convergence risk (Pearson $r = -0.431$, $p = 0.0018$; Spearman $\rho = -0.742$, $p < 0.001$). The stronger Spearman correlation suggests this relationship is robust to outliers and holds across the rank ordering of cases.

WL2018-05 exhibits convergent validity across both frameworks: the lowest accountability openness score (0.039) and the highest PAC convergence risk (1.0), providing case-level evidence that structurally narrow accountability graphs co-occur with premature investigative closure.

Case Clustering

Figure 6 presents K-means clustering ($k = 3$) of cases by graph structural features, projected onto two principal components (PC1: 83.3% variance explained). Three distinct cluster types emerge:

- **Open** (n=16): high violation count, high CFR diversity, openness score ≥ 0.674
- **Standard** (n=33): moderate violation count, moderate openness, openness score 0.373–0.665
- **Extreme convergence** (n=1, WL2018-05): single violation, zero CFR diversity, openness score 0.039

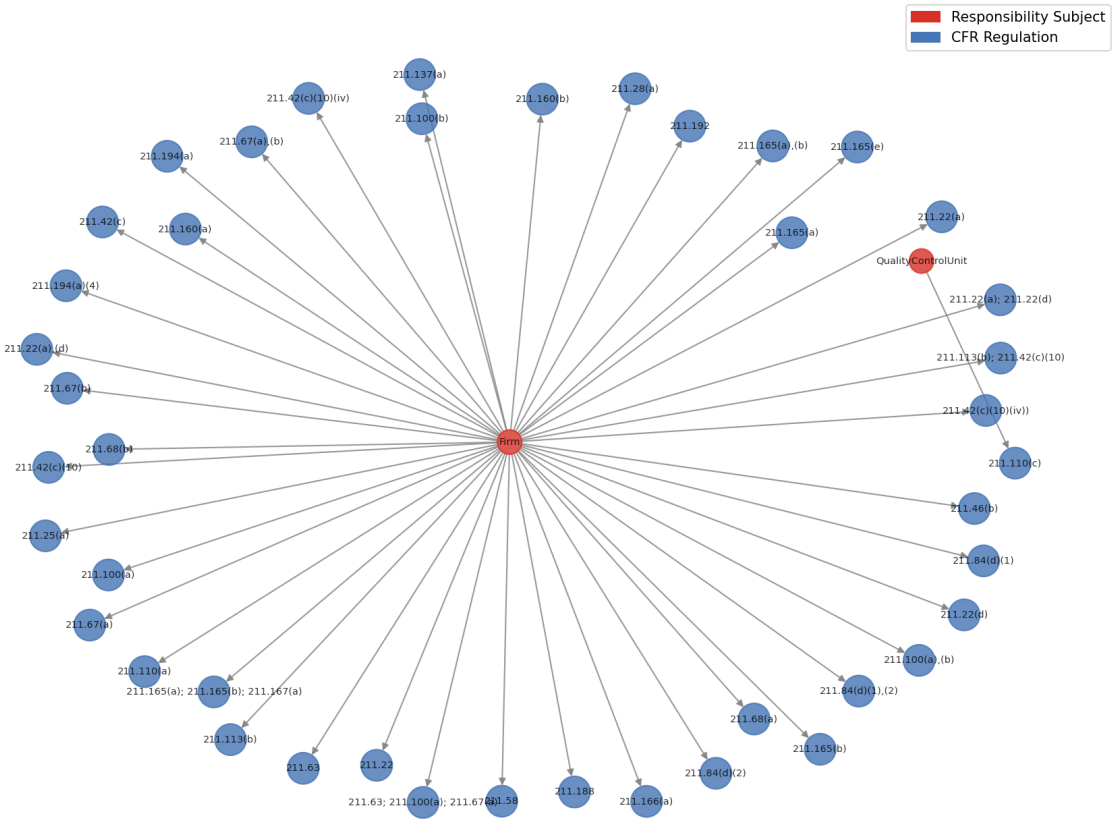


Figure 1: Accountability graph extracted from FDA Warning Letter corpus. Red nodes = responsibility subjects; blue nodes = CFR regulations. Star topology with Firm as dominant hub is evident.

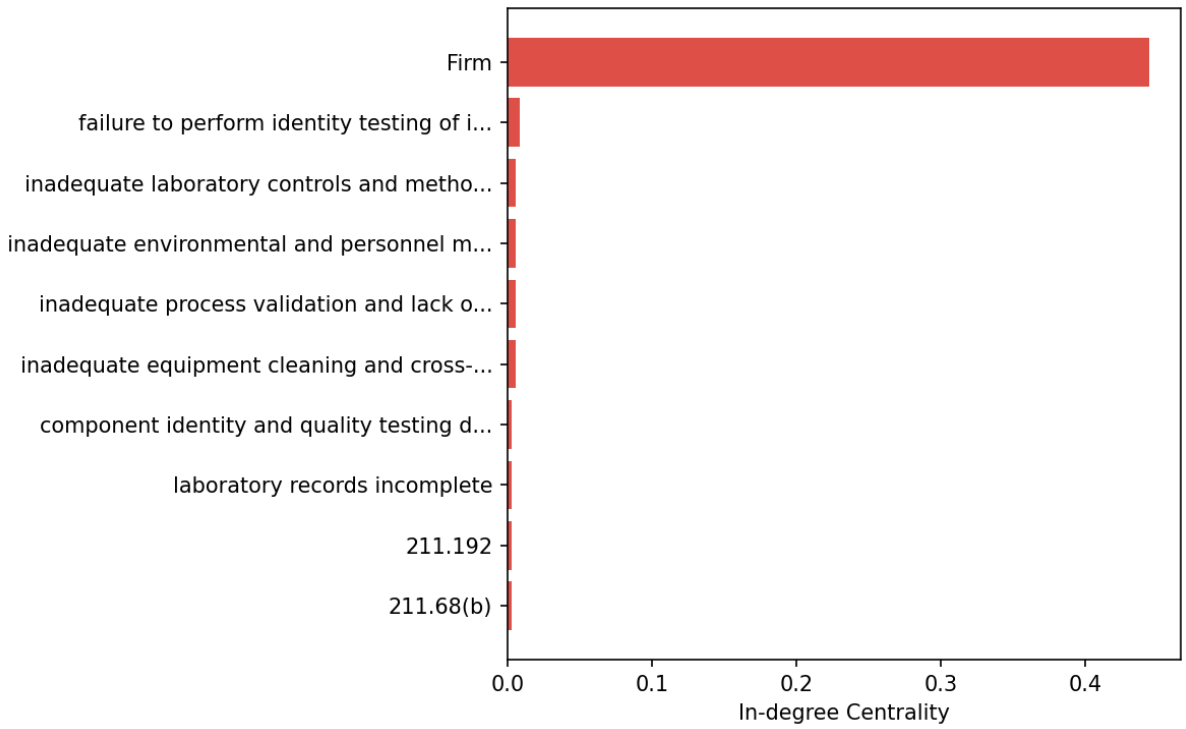


Figure 2: In-degree centrality distribution. Firm exhibits centrality of 0.444, with all other nodes near zero.

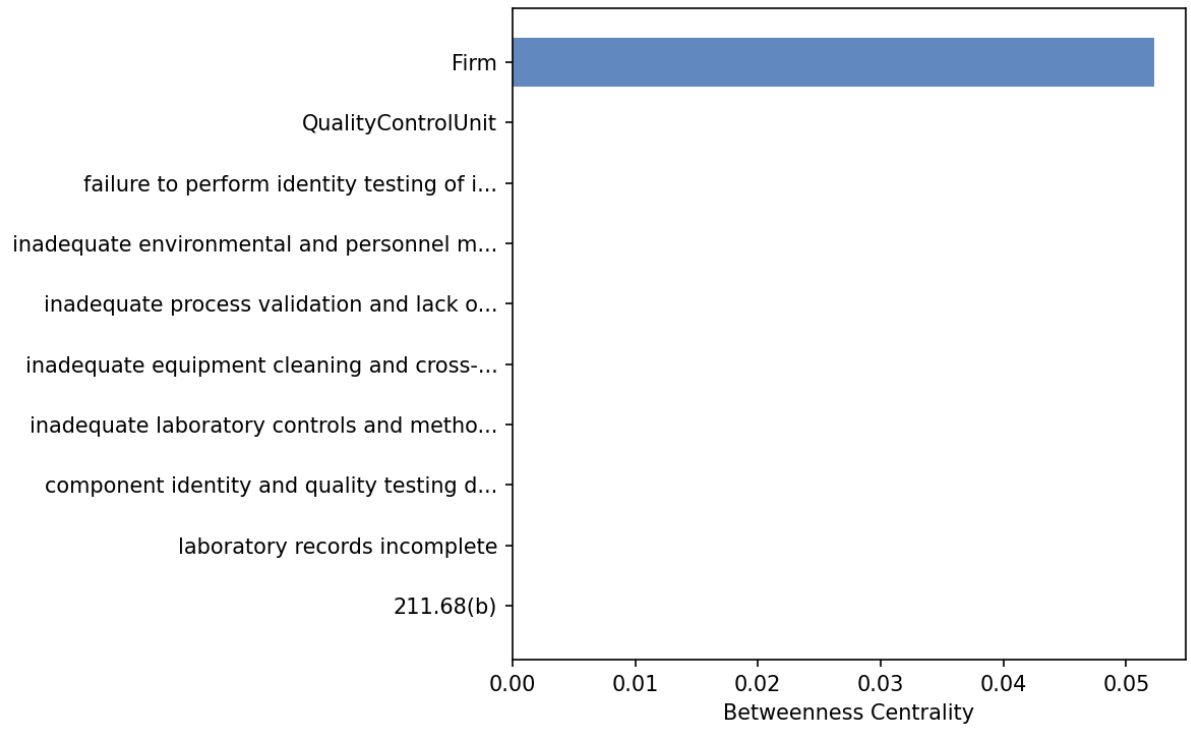


Figure 3: Betweenness centrality distribution. Firm is the sole structural bridge in the accountability network.

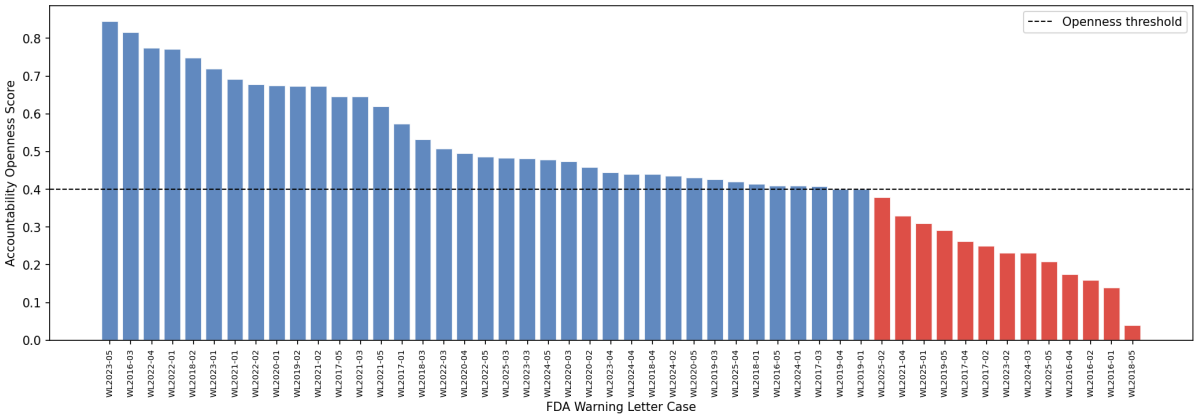


Figure 4: Accountability openness scores across 50 FDA Warning Letter cases. Red = below threshold (concentrated); Blue = above threshold (open).

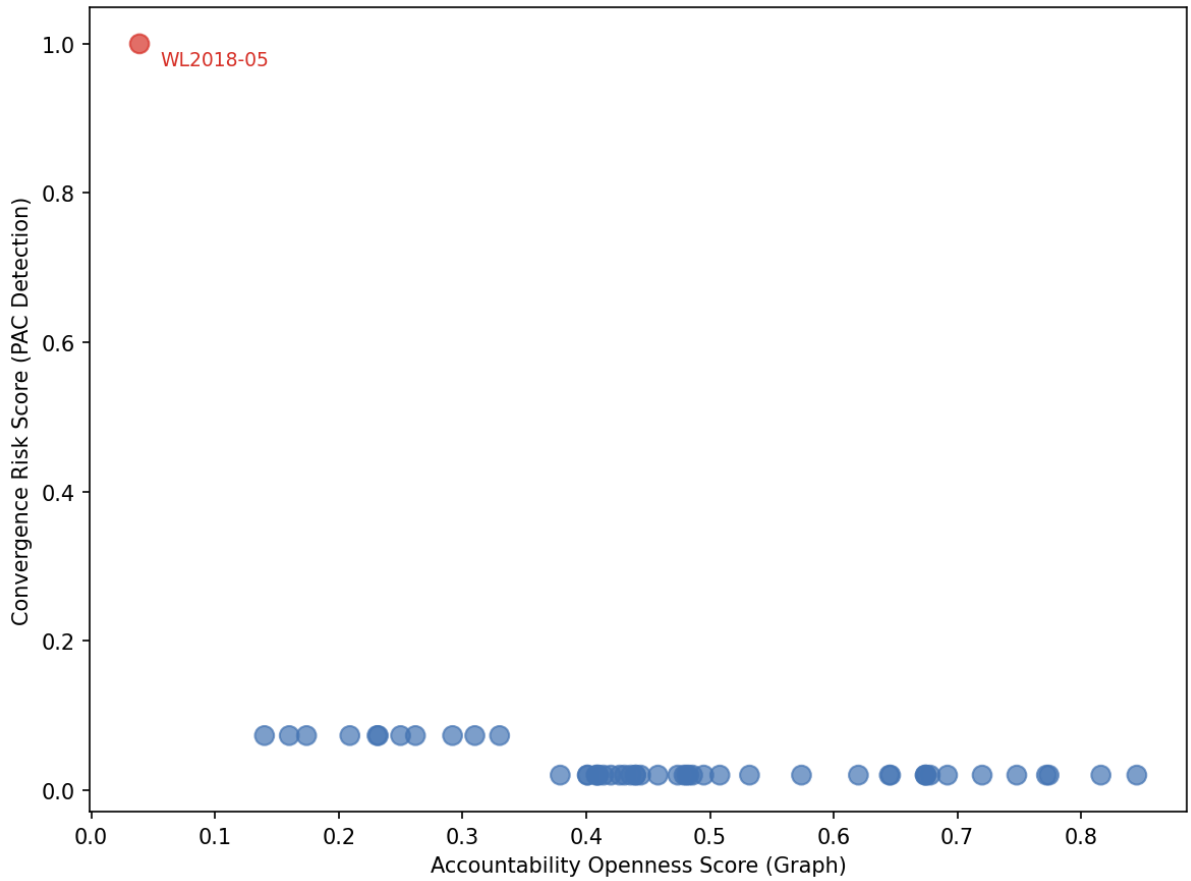


Figure 5: Cross-framework validation: accountability openness score vs PAC convergence risk (Spearman $\rho = -0.742$, $p < 0.001$). WL2018-05 is an extreme outlier in both frameworks.

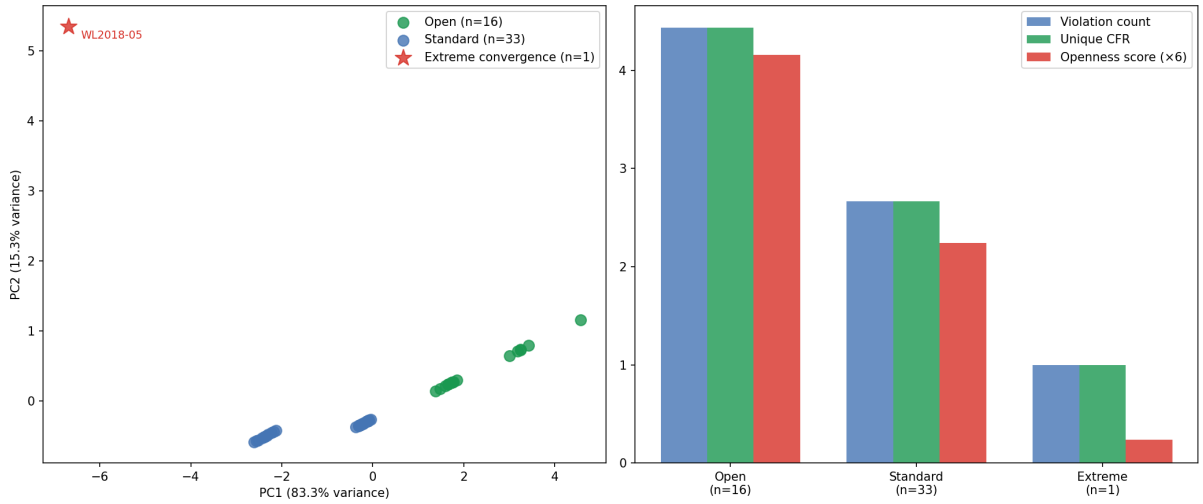


Figure 6: Case clustering by accountability graph structure. Left: PCA projection (PC1 = 83.3% variance). Right: cluster profiles by violation count, CFR diversity, and openness score.

Discussion

Key Findings

Finding 1: FDA Warning Letter accountability graphs exhibit star topology. The graph structure is dominated by a single hub node (Firm), reflecting the enforcement logic of regulatory documents rather than distributed accountability. This structural property is consistent with premature accountability convergence theory (1).

Finding 2: Openness score predicts convergence risk. The significant negative correlation between accountability openness and PAC convergence risk (Spearman $\rho = -0.742$) provides cross-framework validation: cases with structurally narrow graphs are significantly more likely to exhibit premature investigative closure.

Finding 3: Cases cluster into interpretable types. K-means clustering reveals three structurally distinct case types, with WL2018-05 forming a singleton extreme convergence cluster — consistent with its identification as the highest-risk case in the PAC detection framework.

Limitations

Three limitations should be noted. First, the FDA Warning Letter corpus exhibits limited responsibility subject diversity (Firm: 159/160 cases), reflecting the enforcement structure of regulatory documents rather than a limitation of the extraction framework itself. The star topology and near-zero clustering coefficient are corpus-specific properties that may not generalize to other investigative document types. Second, the FDA Warning Letter corpus represents final regulatory outputs rather than in-process investigative reasoning logs, limiting the ability to observe accountability structures as they evolve over the course of an investigation. Third, the accountability openness score uses fixed weights ($w_1 = 0.3$, $w_2 = 0.4$, $w_3 = 0.3$); future work should explore learned weight optimization.

Implications

The graph extraction framework provides a foundation for accountability-aware document analysis systems. By representing accountability as a network rather than a label, it becomes possible to detect structural risk patterns — such as premature convergence toward a single responsible party — before investigative closure occurs.

Conclusion

I presented a graph-based framework for extracting and analyzing accountability structures from regulatory text. Applied to the FDA Warning Letter corpus, the framework reveals star-topology accountability graphs, a significant negative correlation between structural openness and convergence risk (Spearman $\rho = -0.742$, $p < 0.001$), and three interpretable case clusters. WL2018-05 emerges as a convergent extreme case across both graph-based and PAC detection frameworks, providing cross-method validation of structural accountability concentration.

Future work will extend this framework to richer investigative corpora, incorporate temporal graph modeling to track accountability evolution, and integrate graph features into the PAC detection pipeline as structural predictors.

References

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